

Each question is worth 5 points for a total of 25 points. The Extra Credit problem can get you up to an additional 5 points.

Problem 1

Advertisements for a certain small car claim that it floats in water. (a) If the car's mass is 900 kg and its interior volume is 3.0 m^3 , what fraction of the car is immersed when it floats? You can ignore the volume of steel and other materials. (b) Water gradually leaks in and displaces the air in the car. What fraction of the interior volume is filled with water when the car sinks?

Problem 2

An experiment is performed in deep space with two uniform spheres, one with mass 50.0 kg and the other with mass 100.0 kg. They have equal radii, $r = 0.20$ m. The spheres are released from rest with their centers 40.0 m apart. They accelerate toward each other because of their mutual gravitational attraction. You can ignore all gravitational forces other than that between the two spheres. (a) Explain why linear momentum is conserved. (b) When their centers are 20.0 m apart, find (i) the speed of each sphere and (ii) the magnitude of the relative velocity with which one sphere is approaching the other. (c) How far from the initial position of the center of the 50.0 kg sphere do the surfaces of the two spheres collide?

Problem 3

An object is undergoing SHM with period 1.200 s and amplitude 0.600 m. At $t = 0$ the object is at $x = 0$ and is moving in the negative x -direction. How far is the object from the equilibrium position when $t = 0.480$ s?

Problem 4

Three pieces of string, each of length L , are joined together end to end, to make a combined string of length $3L$. The first piece of string has mass per unit length μ_1 , the second piece has mass per unit length $\mu_2 = 4\mu_1$, and the third piece has mass per unit length $\mu_3 = \frac{\mu_1}{4}$. (a) If the combined string is under tension F , how much time does it take a transverse wave to travel the entire length $3L$? Give your answer in terms of L , F , and μ_1 . (b) Does your answer to part (a) depend on the order in which the three pieces are joined together? Explain.

Problem 5

At point A , 3.0 m from a small source of sound that is emitting uniformly in all directions, the sound intensity level is 53 dB. (a) What is the intensity of the sound at A ? (b) How far from the source must you go so that the intensity is one-fourth of what it was at A ? (c) How far must you go so that the *sound* intensity level is one-fourth of what it was at A ? (d) Does intensity obey the inverse-square law? What about *sound* intensity level?

Extra Credit:

Horseshoe bats emit sounds from their nostrils and then listen to the frequency of the sound reflected from their prey to determine the prey's speed. A Horseshoe bat flying at speed v_{bat} emits sound of frequency f_{bat} ; the sound it hears reflected from an insect flying toward it has a higher frequency f_{refl} . (a) Show that the speed of the insect is

$$v_{insect} = v \cdot \frac{f_{refl} \cdot (v - v_{bat}) - f_{bat} \cdot (v + v_{bat})}{f_{refl} \cdot (v - v_{bat}) + f_{bat} \cdot (v + v_{bat})}$$

where v is the speed of sound. (b) If $f_{bat} = 80.7$ kHz, $f_{refl} = 83.5$ kHz, and $v_{bat} = 3.9$ m/s, calculate the speed of the insect.